A Personalized Learning Content Recommendation System Using Hybrid Filtering and Knowledge Tracing

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***Abstract*—Most online learning platforms show the same material to every student, whatever their level. Strong students end up revising topics they mastered weeks ago, while weaker students move ahead without fixing the gaps in their basics. This paper presents a personalized learning content recommendation system that tackles both problems together. The system builds a profile for each student from quiz results, academic records, stated interests, learning history and interaction behaviour. A Bayesian Knowledge Tracing module estimates how well the student has mastered each topic and flags the topics with low mastery as weak. A hybrid engine then ranks learning resources by mixing collaborative filtering, content based filtering and a remediation score that lifts material on weak topics. We describe the architecture and the models behind each module, and we test the design in a controlled simulation of four hundred students. In that simulation, the tracing module recovered truly weak topics with an F1 score of 0.89. The full hybrid placed resources on weak topics in 98 percent of top ten slots, against 35 percent for a preference only hybrid, and the share of weak topics that simulated students went on to master rose from 30 percent to 65 percent. The price was lower precision on predicting what students would open by choice, a trade off that we discuss openly.**

***Index Terms*—personalized learning, recommender systems, knowledge tracing, hybrid filtering, student modelling, e-learning**

I. Introduction

Online education has given students more material than anyone could work through; a single course may carry hundreds of videos, readings and practice sets. The way this material reaches students, though, has barely changed. Most platforms show every enrolled student the same sequence, in the same order and at the same pace. An instructor fixes that sequence once, and it stays fixed for everyone.

This creates two problems that mirror each other. Students who already understand a topic spend time on it again, because the platform has no way of knowing they have mastered it. Students who are falling behind get no extra help, for exactly the same reason. The first group gets bored, the second gets discouraged, and both feed the high dropout rates of online courses.

Streaming services and online shops solved a related problem years ago by ranking items for each user. Moving that idea into education is not a direct copy, however. A movie recommender only has to guess what a user will enjoy. An educational recommender also has to work out what a student *needs*, and that is often the material the student would rather avoid. Someone who is weak in trigonometry seldom clicks on trigonometry revision by choice, so a system trained only on clicks learns to keep it out of sight. A useful educational recommender therefore has to combine preference signals with a separate estimate of what the student actually knows.

This paper proposes a system that makes that combination explicit. Our main contributions are as follows.

- A complete architecture that links four modules: data collection, learner profiling, weakness detection through knowledge tracing, and a hybrid recommendation engine.
- A scoring model that merges collaborative filtering, content based filtering and a remediation term driven by per topic mastery estimates, with weights an institution can tune.
- A weakness detection method built on Bayesian Knowledge Tracing that updates after every quiz answer, so the system reacts within a study session instead of waiting for end of term marks.
- A simulation study that measures how each component changes ranking accuracy, coverage of weak topics and simulated learning gain, along with a sensitivity analysis of the remediation weight.

Section II reviews related work. Section III describes the system and Section IV gives its mathematical formulation. Section V reports the simulation study, and Section VI concludes.

II. Related Work

Work on educational recommenders has grown steadily, and several recent surveys map the field. Khanal et al. reviewed machine learning based recommenders for e-learning and grouped them into content based, collaborative, knowledge based and hybrid families. They found that cold start, sparse data and scalability come up again and again as obstacles [1]. Raj and Renumol studied adaptive content recommenders in personalized learning environments from 2015 to 2020 and

observed that most working systems end up hybrid, because no single technique copes with every situation [2]. Reviews by Dhananjaya et al. [3] and Bin et al. [4] confirm a move toward machine learning and deep learning. Bin et al. also look closely at how learners and learning objects are modelled, and in our reading of the systems they cover, the learner is still often described very thinly, sometimes as little more than a vector of past ratings.

The building blocks are well known. Collaborative filtering grew up in commercial settings, and matrix factorization became the standard latent factor method after the work of Koren, Bell and Volinsky [5]. Content based filtering comes from classical information retrieval, where the TF-IDF weighting of Salton and Buckley is still the usual way to turn text into comparable vectors [6]. Burke catalogued the ways these methods can be combined, from simple weighted mixing to cascade and switching designs [7]. Herlocker et al. set out the evaluation practice for recommenders, including the ranking measures we use later [8].

On the student side, the central idea is knowledge tracing. Corbett and Anderson introduced Bayesian Knowledge Tracing (BKT) in 1994. It treats mastery of each skill as an unobserved yes or no state and revises its belief about that state after every answer [9]. Piech et al. replaced the hand set probabilities with a recurrent neural network in Deep Knowledge Tracing (DKT) and reported clear gains in predicting future answers [10]. Attention based models such as SAKT came next [11]. The survey by Abdelrahman et al. follows this line from early probabilistic models to memory based and attention based networks [12]. A recurring point in this literature is that the deep models gain accuracy but lose the easy interpretation that BKT offers, which matters when teachers need to trust the output. Item Response Theory gives a static view of ability and item difficulty [13].

What is still uncommon is a tight link between knowledge tracing and the recommendation step itself. Many systems use mastery estimates only on dashboards, and many recommenders use only preference data. Wu et al. describe student modelling and personalized recommendation as the two pillars of personalized learning and study how they interact [14]. In many working systems, though, the two remain only loosely connected. Our design is built around closing that gap: the mastery estimates feed straight into the ranking function as a scoring term of their own.

III. Proposed System

A. Overview

The system sits between an existing learning management system (LMS) and the student. Its four modules form a loop, as shown in Fig. 1. The engine produces a ranked list of resources, the student reacts to that list, and the reaction flows back into data collection.

B. Data Collection

The data collection layer gathers five kinds of signal. *Academic performance* covers marks from assignments, internal tests and earlier semesters, imported from the LMS gradebook. *Quiz results* record, for each question, whether the answer was right, how many attempts it took and how long the student spent. *Interests* are the topics and content formats a student picks during onboarding, such as a preference for video over text. *Learning history* logs which resources were opened, finished, bookmarked or abandoned. *Interaction behaviour* holds finer events such as pause and rewind points in videos, scroll depth in readings and session length. Every question and resource must carry at least one topic tag, which is the one manual task the system asks of content authors.

C. Learner Profiling

The profiling module condenses the raw signals into three parts. The first is a preference vector over content features such as format, difficulty and topic area. The second is a student by resource engagement matrix used by collaborative filtering. The third is the per topic answer history, which is passed to weakness detection. Profiles are refreshed after every session, not in a nightly batch, so the system can react to a failed quiz before the next session.

D. Weakness Detection

For each topic, the module runs BKT over the student's sequence of answers and keeps a current mastery probability. A topic is flagged weak when that probability falls below a threshold θ, which can be set per course. We use 0.6 by default. This is well under the 0.95 criterion common in tutoring systems, so help arrives before a gap becomes serious. The module also separates "weak" from "unseen". A topic with too few answers is treated as unknown, which stops a new student from being flooded with remedial material on the first day. Fig. 2 shows the underlying two state model.

E. Hybrid Recommendation Engine

The engine gives every candidate resource three scores. A collaborative score estimates how students with similar habits engaged with it. A content score measures how well its features match the student's preference vector. A remediation score lifts resources tagged with the student's weak topics, in proportion to how weak each topic is. The three scores are normalised and mixed with weights that sum to one, as shown in Fig. 3.

The top results are shown in two labelled groups: "Recommended for you" and "Strengthen your basics". We chose to keep the remedial group visible but separate on purpose. If remedial items are mixed silently into the general list, students tend to skip past them. A labelled group with a short reason attached, for example "you scored 40 percent on recent trigonometry questions", gives the student a reason to engage.

F. Feedback Loop

Each recommendation shown is logged with the student's reaction: opened, completed, ignored or dismissed. Completions raise the engagement values used by collaborative filtering. A resource that keeps being ignored slowly loses its score

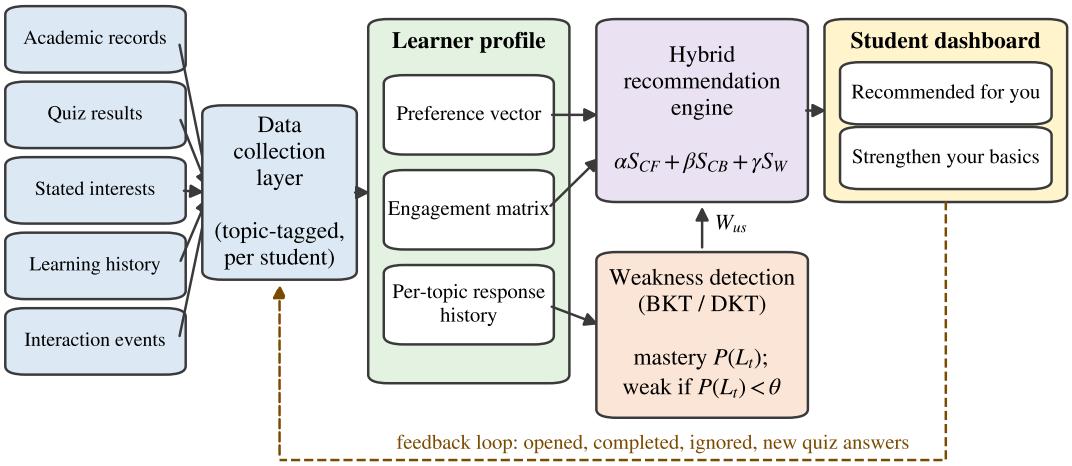


Fig. 1. Architecture of the proposed system. Signals from the LMS are profiled, weak topics are detected by knowledge tracing, and the hybrid engine fills two labelled groups on the student dashboard. The dashed path is the feedback loop.

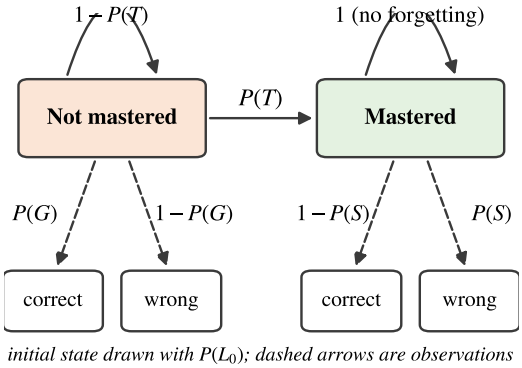


Fig. 2. Two state Bayesian Knowledge Tracing model for one topic, with learning, guess and slip probabilities.

Algorithm 1 One recommendation round for student u

Require: profile of u, BKT parameters, weights α, β, γ, threshold θ, list size K

for each topic s with at least $n_{\min}$ answers **do**
  update $P(L_t)_{us}$ with (7) to (9)
  $W_{us} \leftarrow 1 - P(L_t)_{us}$
end for
for each unseen resource i **do**
  compute $S_{CF}(u,i)$, $S_{CB}(u,i)$, $S_W(u,i)$ and normalise
  $S(u,i) \leftarrow \delta^{g(u,i)} \left[\alpha S_{CF} + \beta S_{CB} + \gamma S_W\right]$
end for
show top K by S as "Recommended for you"
show top K with $S_W > 0$ as "Strengthen your basics"
log reactions and new answers for the next round

for that student, which keeps the list fresh. New quiz answers on a weak topic go straight back to the tracing module. If the topic is still weak after remedial study, the system keeps it in the remedial group but serves different material, since showing the same resource again after it failed to help rarely works. Algorithm 1 summarises one recommendation round.

IV. Mathematical Formulation

A. Content Based Filtering

Each resource i is described by a feature vector $\mathbf{v}_i$ built from its topic tags, difficulty, format and the TF-IDF weights of its text description. The TF-IDF weight of term t in description d is

$$w_{t,d} = tf_{t,d} \cdot \log \frac{N}{df_t}, \tag{1}$$

where $tf_{t,d}$ is the term frequency, N is the number of resources and df_t is the number of resources that contain t [6]. The preference vector $\mathbf{p}_u$ of student u is the engagement weighted mean of the vectors of resources the student has completed. The content score is the cosine similarity

$$S_{CB}(u,i) = \frac{\mathbf{p}_u \cdot \mathbf{v}_i}{\|\mathbf{p}_u\| \, \|\mathbf{v}_i\|}. \tag{2}$$

B. Collaborative Filtering

Explicit ratings are rare in education, so the matrix R holds implicit engagement scores r_{ui} on a scale of 1 to 5, derived from completion and revisits. For small cohorts we use a neighbourhood method. The similarity of students u and v is the Pearson correlation, where every sum runs over the set I_{uv} of resources both have used,

$$\mathrm{sim}(u,v) = \frac{\sum_i (r_{ui} - \bar{r}_u)(r_{vi} - \bar{r}_v)}{\sqrt{\sum_i (r_{ui} - \bar{r}_u)^2} \sqrt{\sum_i (r_{vi} - \bar{r}_v)^2}}, \tag{3}$$

and the predicted engagement with an unseen resource is

$$\hat{r}_{ui} = \bar{r}_u + \frac{\sum_{v \in N_k(u)} \mathrm{sim}(u,v) \, (r_{vi} - \bar{r}_v)}{\sum_{v \in N_k(u)} |\mathrm{sim}(u,v)|}, \tag{4}$$

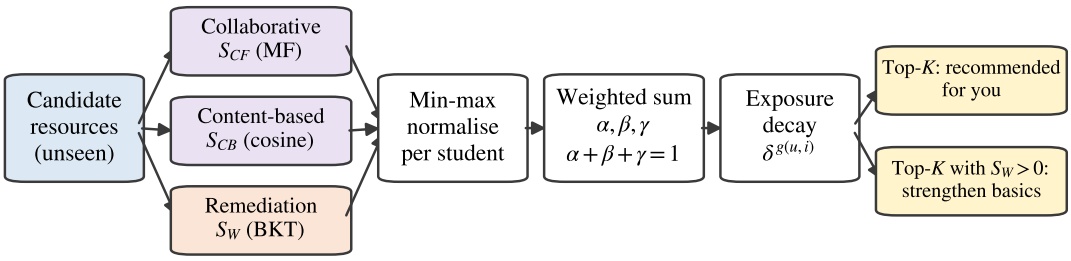


Fig. 3. Hybrid scoring pipeline. Three scorers are normalised, combined with weights, adjusted for repeated exposure and split into the two dashboard groups.

where $N_k(u)$ holds the k most similar students. For larger cohorts we switch to biased matrix factorization [5], which learns latent vectors $\mathbf{x}_u$, $\mathbf{y}_i$ and biases b_u, b_i by minimising

$$\min_{\mathbf{x},\mathbf{y},b} \sum_{(u,i)\in\mathcal{K}} \left(r_{ui} - \mu - b_u - b_i - \mathbf{y}_i^\top \mathbf{x}_u\right)^2 + \lambda \left(\|\mathbf{x}_u\|^2 + \|\mathbf{y}_i\|^2 + b_u^2 + b_i^2\right), \tag{5}$$

where $\mathcal{K}$ is the set of observed pairs, μ is the global mean and λ controls regularisation. The prediction is $\hat{r}_{ui} = \mu + b_u + b_i + \mathbf{y}_i^\top \mathbf{x}_u$. In both cases $S_{CF}(u,i)$ is $\hat{r}_{ui}$ rescaled to $[0,1]$.

C. Weakness Detection with Knowledge Tracing

For each topic s, BKT uses four parameters: initial mastery $P(L_0)$, learning rate $P(T)$, slip $P(S)$ and guess $P(G)$ [9]. Given the current estimate $P(L_t)$, the chance of a correct answer is

$$P(C_t) = P(L_t)\left(1 - P(S)\right) + \left(1 - P(L_t)\right) P(G). \tag{6}$$

After a correct answer the estimate is revised by Bayes' rule,

$$P(L_t \mid C) = \frac{P(L_t)\left(1 - P(S)\right)}{P(L_t)\left(1 - P(S)\right) + \left(1 - P(L_t)\right) P(G)}, \tag{7}$$

and after a wrong answer,

$$P(L_t \mid \bar{C}) = \frac{P(L_t)\, P(S)}{P(L_t)\, P(S) + \left(1 - P(L_t)\right)\left(1 - P(G)\right)}. \tag{8}$$

The estimate then moves forward by one learning opportunity,

$$P(L_{t+1}) = P(L_t \mid \text{obs}) + \left(1 - P(L_t \mid \text{obs})\right) P(T). \tag{9}$$

The weakness score of student u on topic s is

$$W_{us} = 1 - P(L_t)_{us}, \tag{10}$$

and topic s is flagged when $P(L_t)_{us} < \theta$ and the student has at least $n_{\min}$ answers on it. Once enough data has built up, DKT [10] can replace BKT. DKT feeds one hot (topic, correctness) pairs $\mathbf{z}_t$ through a recurrent network,

$$\begin{aligned} \mathbf{h}_t &= \tanh(W_{hz}\mathbf{z}_t + W_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h), \\ \mathbf{o}_t &= \sigma(W_{oh}\mathbf{h}_t + \mathbf{b}_o), \end{aligned} \tag{11}$$

where $\mathbf{o}_t$ holds the predicted chance of answering each topic correctly at the next step. The definition in (10) stays the same, with $P(L_t)_{us}$ replaced by the matching entry of $\mathbf{o}_t$.

D. Hybrid Ranking with Remediation

Let $T(i)$ be the topic tags of resource i. Its remediation score is the strongest weakness it addresses,

$$S_W(u,i) = \max_{s\in T(i)} W_{us}\, \mathbb{I}[P(L_t)_{us} < \theta], \tag{12}$$

where $\mathbb{I}[\cdot]$ is the indicator function. The final score is

$$S(u,i) = \delta^{\,g(u,i)} \left[\alpha\, S_{CF} + \beta\, S_{CB} + \gamma\, S_W\right], \tag{13}$$

with all three scores taken at (u,i) and $\alpha + \beta + \gamma = 1$. Here $g(u,i)$ counts how often resource i was shown to u without being opened, and $\delta \in (0,1)$ sets how fast ignored items fade; we use $\delta = 0.85$. Our default weights are $\alpha = 0.4$, $\beta = 0.3$ and $\gamma = 0.3$. For a new student with no history, we set $\alpha = 0$ and rely on content matching against onboarding interests, the usual way of easing cold start [1].

E. Evaluation Measures

Ranking quality is measured with the usual top K measures [8]. With Rel_u the held out resources of u and Rec_u^K the top K list,

$$\mathrm{P@}K = \frac{|\mathrm{Rel}_u \cap \mathrm{Rec}_u^K|}{K}, \tag{14}$$

$$\mathrm{NDCG@}K = \frac{1}{\mathrm{IDCG@}K} \sum_{j=1}^{K} \frac{2^{\,\mathrm{rel}_j} - 1}{\log_2(j+1)}, \tag{15}$$

where rel_j is the engagement score of the item at rank j divided by 5, and zero if the item is not in Rel_u. Prediction error on held out engagement scores is measured by

$$\mathrm{RMSE} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{(u,i)\in\mathcal{T}} (\hat{r}_{ui} - r_{ui})^2}, \tag{16}$$

over the test set $\mathcal{T}$, with mean absolute error (MAE) defined in the same way. Because P@K only rewards items a student would have opened anyway, we add a measure of remedial value. With W_u^* the set of topics in which student u is truly weak,

$$\mathrm{Cov@}K = \frac{1}{K} \sum_{i\in\mathrm{Rec}_u^K} \mathbb{I}[T(i) \cap W_u^* \neq \emptyset]. \tag{17}$$

Knowledge tracing is judged by the area under the ROC curve (AUC) for predicting the next answer, which is standard in that literature [10].

V. Simulation Study

A. Why Simulation

Public datasets such as ASSISTments [15] and EdNet [16] suit knowledge tracing well, but they cannot tell us which topics a student was truly weak in, or what would have happened under different recommendations. We therefore built a controlled simulation in which the true mastery of every student is known. This lets us measure coverage of real weaknesses and run what if experiments on learning. The results describe how the method behaves under our stated assumptions. They are not evidence of real classroom gains, which need the pilot study outlined in Section VI.

B. Setup

We generated 400 students, 10 topics and 300 resources. Each resource carries one or two topic tags, one of three formats and one of three difficulty levels. Each student has a hidden mastered or not mastered state per topic, drawn with probability $P(L_0) \sim U(0.25, 0.75)$, and true BKT parameters were drawn per topic with $P(T) \sim U(0.05, 0.20)$, $P(S) \sim U(0.05, 0.15)$ and $P(G) \sim U(0.15, 0.30)$. Every student answered between 4 and 11 quiz questions per topic. Students also have a six dimensional latent taste vector and a preference over formats. One assumption matters a great deal: students show positive interest in topics they have mastered and negative interest in topics they have not, which models the avoidance behaviour described in the introduction. Each student then engaged with 40 resources, sampled in proportion to their affinity, with scores from 1 to 5. Interaction sequences were split in order, with the first 80 percent used for training and the last 20 percent for testing.

BKT parameters were fitted per topic by grid search maximum likelihood on 70 percent of the students and tested on the other 30 percent. Matrix factorization used eight latent factors, $\lambda = 2$ and 30 rounds of alternating least squares. The content vector holds topic, format and difficulty, since synthetic resources have no real text to weight with (1).

To estimate learning gain we ran four recommendation rounds. In each round every student saw a top five list. The chance of opening an item followed the student's affinity, with a small bonus of 0.15 for items in the labelled remedial group. An opened item on an unmastered topic moved that topic to mastered with probability 0.25. After each round, two quiz questions per topic updated the BKT estimates. Gain is the share of initially weak (student, topic) pairs that ended up mastered, averaged over five random seeds. In this simulation, items already shown are never shown again, so the decay term in (13) is not exercised.

C. Knowledge Tracing and Weakness Detection

Table I reports the tracing results on the held out students. BKT reached an AUC of 0.816 for predicting the next answer. A simple running average of past correctness came close at 0.805, since simple baselines are hard to beat on short answer sequences. Detection itself was strong. Of the 1,711 truly weak (student, topic) pairs, BKT flagged 81.7 percent, and 98.0 percent of its flags were correct. The method is cautious: it would rather wait a few more questions than send remedial work to a student who does not need it. Fig. 4 shows how the estimate moves for one student who starts badly and then improves.

TABLE I
Knowledge Tracing and Weakness Detection Results

Measure	Value
Next answer AUC, BKT	0.816
Next answer AUC, running average baseline	0.805
Weak topic detection precision	0.980
Weak topic detection recall	0.817
Weak topic detection F1	0.891

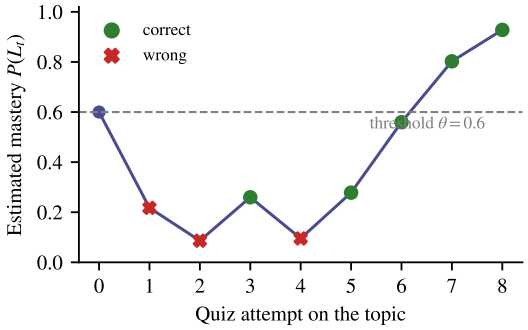


Fig. 4. Mastery estimate for one simulated student on one topic. Early wrong answers flag the topic; a run of correct answers clears the flag.

D. Recommendation Results

The matrix factorization model predicted held out engagement with an RMSE of 0.714 and an MAE of 0.486 on the 1 to 5 scale. Table II compares four configurations at $K = 10$. Adding content matching to collaborative filtering roughly doubled precision and NDCG, which is what one would expect from a preference model. It also made the remedial picture worse. Coverage of weak topics fell from 0.548 to 0.347, and learning gain fell from 0.366 to 0.296. Content matching learns that a student avoids certain topics and dutifully keeps them away, which is exactly the trap described in the introduction.

The full hybrid reverses this. Almost every slot in its top ten list touched a true weakness, and simulated students mastered 65.0 percent of their weak topics, more than twice the share under the preference only hybrid. The cost is a lower P@10, because many recommended items are ones the student would not have chosen alone. We consider this an honest trade off rather than a defect. P@K rewards agreement with past behaviour, and in a learning setting past behaviour is partly the problem being solved.

E. Sensitivity to the Remediation Weight

Fig. 5 varies γ from 0 to 0.6 while keeping α and β in the ratio 4 to 3. Most of the benefit arrives early. Coverage climbs to 0.93 and gain to 0.62 by $\gamma = 0.2$, and both level off after about 0.3. Precision falls over the same range and then also flattens. In practice this means a moderate weight gets nearly

TABLE II
RECOMMENDATION RESULTS AT $K = 10$ (SIMULATION)

Configuration	P@10	NDCG@10	Cov@10	Gain
Popularity	0.034	0.041	0.392	n/a
CF only	0.043	0.046	0.548	0.366 ± 0.013
CF + CB ($\gamma = 0$)	**0.084**	**0.102**	0.347	0.296 ± 0.006
Full hybrid ($\gamma = 0.3$)	0.046	0.050	**0.980**	**0.650** ± **0.010**

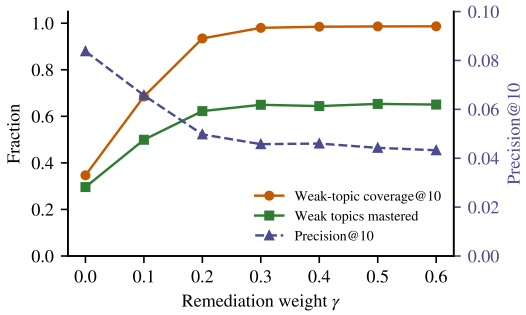


Fig. 5. Effect of the remediation weight on coverage, learning gain and precision.

all of the remedial value, and there is little reason to push γ higher. Institutions that care more about engagement can choose a value near 0.1 and still double weak topic coverage compared with no remediation at all.

F. Limitations

Several limits should be kept in mind. The size of the gain depends on our assumptions, especially the avoidance of weak topics and the chance that a resource teaches its topic, and real values will differ. Synthetic resources have no text, so the TF-IDF part of the content score was not tested. The small open bonus for labelled remedial items is a guess about how students respond to an explained recommendation; a real pilot is needed to measure it. Finally, the BKT fitting used a coarse grid, and finer fitting by expectation maximisation may change the detection figures slightly.

VI. CONCLUSION AND FUTURE WORK

This paper presented a personalized learning content recommendation system that treats remediation of weak topics as part of the ranking function instead of a separate dashboard feature. It combines familiar parts, namely content matching, collaborative filtering and Bayesian Knowledge Tracing, in one scoring formula with weights that are easy to explain. In a controlled simulation, the tracing module found weak topics with an F1 score of 0.89, and the full hybrid more than doubled the share of weak topics that students went on to master compared with a preference only design. The simulation also showed why the remediation term is needed: a pure preference model learns student avoidance and makes the problem worse.

Several directions remain open. Topic tagging puts real work on content authors, and automatic tagging with language models is a natural next step. BKT can be replaced by attention based tracing once course data grow large enough. The current engine ranks single resources, while weak topics often need an ordered path through prerequisite material, so learning path recommendation is a clear extension. We also plan to evaluate the tracing and collaborative modules on ASSISTments [15] and EdNet [16]. Most importantly, we plan a classroom pilot at our institution in which one section follows the usual fixed sequence and another uses the system. The main outcome will be change in quiz accuracy on flagged topics over two weeks, with open rates of remedial items as a secondary measure. Interaction logs will be pseudonymised at collection, and consent will be obtained before any data are used.

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